

Decimal Approximation + Application

Unit 4 · decimal operations and approximation · 65 minutes · 1-to-3 Online Lesson

Corresponding textbooks: "New Thinking in Primary School Mathematics (Second Edition)" Volume 5, Volume A, Unit 4 + Modern Education, Volume 5, Unit 11

Core traps: ☐ T2 Approximate value positioning - count one more digit before judging · Early braking

SSPA Related:  IF Common in paper 2, estimation and approximation are compulsory questions

Prerequisite knowledge: Class 12 (Decimal multiplication · Decimal place processing · Decimal point alignment)

Objective of this class: ❶ Master rounding to tenths/percentiles ❷ Understand "count one more digit before making a judgment" ❸ The difference between estimation and approximation ❹ Solve problems with applications

Student Name:

Class:

Date:

Time Spent:

I.Warm-Up Questions (total 5 question , 5 minutes)

#	Question	Difficulty	Working Space (write out complete process)
1	Read 3.27 as: _____; where "2" is in the _____ position and the value is _____.	Basic	
2	Compare size (fill in >, < or =): (a) 0.7 _____ 0.70 (b) 1.05 _____ 1.5 (c) 2.30 _____ 2.3	Basic	
3	Multiply 4.5 by 10 = _____; multiply 4.5 by 100 = _____. Do you notice how the decimal point moves?	Basic	
4	Calculation: $3.6 + 2.48 = ?$ (Pay attention to aligning the decimal points)	Advanced	
5	Calculation: $5.2 \times 0.3 = ?$ (For decimal multiplication, count as an integer first and then add the decimal point)	Advanced	

II.Core Knowledge + Worked Examples

Knowledge point 1: Approximate decimal values - round to tenths/percentiles 🟡 SSPA

Rounding method:

- ① **Look at the next digit of the target bit |||SEP|||: Get to tenth place → Look at hundredth place; Get to hundredth place → Look at thousandth place** Judgment rules |||SEP|||: Next digit $\geq 5 \rightarrow$ Target digit +1 (carry); Next digit $\leq 4 \rightarrow$ Target digit remains unchanged (removed)
- ② **Key tips |||SEP|||: "Count one more digit before making a judgment, so as not to stop early!"** Note |||SEP|||: The last zero of the approximate answer must be retained (for example, $3.20 \neq 3.2$, the former is accurate to the percentile)
- ③ **Key formula:** "Calculate one more person before making a judgment, so as not to brake early!"
- ④ **Notice:** The last zeros of the approximate answer must be retained (for example, $3.20 \neq 3.2$, the former is accurate to the hundredth)

❑ Example of trap detonation (the most important demonstration in this class)

Round 4.368 to the hundredth (that is, to two decimal places).

✗ Common mistakes (braking early)

4.36 (I only saw 8 thousandths, but my judgment was wrong)
Or write 4.37 directly (carry randomly)

Have not mastered the "look at the last digit" rule: to get the percentile, look at the thousandth place (the 3rd place), 4.368
→ the thousandth place is $8 \geq 5 \rightarrow$ the percentile $6+1=7$

✓ Correct solution

$$4.368 \approx 4.37$$

Take the percentile (2nd place) = 6, and look at the thousandth place (3rd place) = 8 ≥ 5 , so the percentile is rounded up by 1
→ 4.37

 **Tip: "Look behind the target position. If it is five or more, carry it in. If it is four or less, discard it. Keep the last zero!"**

 **The most frequent error: when calculating the tenth place, only look at the ones digit and forget to look at the hundredth place. Remember: always look at the "right one" of the target position!**

 **The second most frequent error: the last 0 bits of the approximation result are removed. For example, 2.097 $\approx$ 2.10 (percentile) cannot be written as 2.1!**

Knowledge Point - Worked Examples (write out determine process: objective bit $\rightarrow$ which digit to look at $\rightarrow$ the digit $\rightarrow$ rounding/rounding $\rightarrow$ answer)

#	Question	Difficulty	Working Space
Example 1	Round 2.73 to the tenth place (i.e. 1 decimal place).		
Example 2	Round 5.084 to the hundredth (that is, 2 decimal places).		

Knowledge point 2: Application of decimal approximations - amount, length, weight SSPA required test

Common application scenarios:

- ① **Amount** |||SEP|||: Hong Kong dollars are rounded to cents (tenths) or cents (percents) $\rightarrow$ Amounts in daily life are usually rounded to two decimal places
- ② **Length** |||SEP|||: Meters are rounded to cm (hundredths) or mm (thousandths)
- ③ **Weight**: Kilogram to the nearest gram (thousandth place)
- ④ **Key judgment**: Confirm the unit requirements first, and then decide which digit to get
- ④ **critical judgment**: Check the unit requirements first, and then decide which one to get.

①

Recognize unit

Which position does the question require?

②

Target position

Tenth place? Percentile?

③

Look at the last one

Determine carry or discard

④

write answer

Keep units and trailing zeros

example

Example 3: A rope is 3.467 m long, approximated to tenths (that is, m is taken to one decimal place).

example

Example 4: A bag of rice weighs 2.385 kg, rounded to the nearest percentile (i.e. kg is rounded to two decimal places).

Knowledge Point 2 Synchronization practice

#	Question	Difficulty	Working Space
6	A pencil is 18.63 cm long, rounded to the nearest tenth.		
7	A book weighs 0.856 kg, rounded to the nearest percentile.		
8	Water bottle capacity 1.275 L, rounded to the nearest percentile.		
9	An orange weighs 0.204 kg, rounded to the nearest tenth.		
10	A road is 2.951 km long, approximated to the nearest tenth.		

Knowledge point 3: The difference between estimation and approximation 🟡 SSPA Advanced

Estimation $\neq$ Approximation:

- ① **Estimation |||SEP|||:** Use "approximation" instead of an exact number for fast calculation, such as $3.8 \times 2.1 \approx 4 \times 2 = 83.85$ (percentile)
- ② **The estimate is used "before calculation"** (estimate first and then calculate),
- ③ **The approximate value is used "after calculation"** (the position is calculated after calculation)
- ④ **Common test for sub-tests: first estimate to test the rationality of the answer $\rightarrow$ then calculate accurately $\rightarrow$ finally take the approximate value** (Take your seat after counting)
- ④ Sub-test common test: first estimate and test the rationality of the answer $\rightarrow$ then calculate accurately $\rightarrow$ finally take the approximate value

example

Example 5: What is the approximate result of estimating 4.7×3.2 ? Then calculate it accurately, and then approximate the answer to tenths.

example

Example 6: Xiao Ming said "The approximate value of 3.14159 is 3.1 (tenths)", Xiao Hua said " $3.14159 \approx 3$ estimates". Whose statement is correct? Why?

Knowledge Point 3 Synchronization practice

#	Question	Difficulty	Working Space
11	Estimate $5.8 \times 4.1 \approx \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$		
12	Estimate $9.7 + 3.2 + 6.1 \approx \underline{\hspace{1cm}} + \underline{\hspace{1cm}} + \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$		

III. Lesson Layered Synchronization Practice

Basic layer (total 5 questions, everyone must do)

#	Question	Difficulty	Working Space
13	Round 7.82 to the tenth place.		
14	Approximate 1.456 to the percentile.		
15	Round 0.395 to the percentile.		
16	Round 4.072 to the nearest tenth place.		
17	Round 9.998 to the percentile.		

Advanced layer (total 5 questions,   choose do)

#	Question	Difficulty	Working Space
18	Approximate 6.2549 to the percentile.		
19	Round 3.097 to the nearest tenth place. Should I keep the last zero in my answer? Why?		
20	An item is priced at \$12.875. The supermarket rounds the price to the nearest tenth. How much should be charged?		
21	A fish weighs 0.683 kg. Approximate values: (a) to tenths (b) to hundredths. What's the difference between the two answers?		
22	To calculate 3.25×1.6 , first estimate (round up), then calculate exactly, and finally round the exact answer to the nearest tenth.		

 challenge layer (total 3 questions,  choose do, SSPAKiller Questions)

#	Question	Difficulty	Working Space
23	Approximate 0.9999: (a) to tenths (b) to hundredths (c) to thousandths. Observing the three answers, what did you find?		
24	To calculate $4.56 + 7.389 + 2.1$, first approximate each number to the tenth place and add them together. Is the result the same as adding first and then approximating to tenths?		
25	A rectangle is 4.37 cm long and 2.86 cm wide. (a) Calculate the area (to the nearest hundredth). (b) First, approximate the length and breadth to tenths, and then calculate the area. What is the difference between the two areas?		

Knowledge point 4: Approximate value application questions (subject to sub-test text questions) 🟡 SSPA compulsory exam

Four steps to solve the problem: ① Circle keywords ("approximately" "approximately" "get to") ② Recognize the positioning requirements ③ Accurate calculation → then approximate ④ Write the complete answer with the units

example

Example 7: Xiao Ming bought 3 books, the prices are \$12.85, \$23.49, and \$8.76 respectively. How much did he pay in total? (Answers are approximated to tenths, i.e. rounded to millimeters)

applicationquestionpractice (all must do, column → calculate → approximate → answer sentence)

#	Question	Difficulty	Working Space
26	A rope is 5.738 m long. Approximately how many meters long is the rope, approximated to the nearest hundredth?		
27	A pencil costs \$4.75 and a rubber eraser costs \$3.28. How much is it worth in total? (Answers are approximated to tenths)		
28	A kettle can hold 1.863 L of water. Approximately how many L of water can the kettle hold?		

advancedapplicationquestion (🧑🏫🚀 choose do, SSPAPaper 2commonquestion type)

#	Question	Difficulty	Working Space
29	The three fruits weigh 0.385 kg, 0.472 kg and 0.298 kg respectively. (a) What is the total weight in kg? (b) Round the total weight to the nearest tenth. (c) If each weight is approximated to the nearest tenth and then added together, will the result be the same as (b)?		
30	The side length of the square is 2.67 cm. Find the perimeter and approximate the answer to the nearest tenth.		
31	The rectangle is 5.38 m long and 2.94 m wide. (a) Find the perimeter, approximating to the nearest tenth. (b) Find the area, approximating to the percentile.		

IV. 🏔️ Ultimate challenge area

#	Question	Difficulty	Working Space
 1	A number rounded to the tenth place is 7.0. What is the smallest possible number? What is the maximum possible? (Tip: Consider the boundaries between "four giving		

	out" and "five in")		
 2	Calculate $0.1 \div 3 = 0.03333\ldots$ (infinite loop). Approximate the result: (a) to tenths (b) to hundredths (c) to thousandths. Write the judgment process for each answer.		
 3	A trapezoid has an upper base 3.26 cm, a lower base 5.74 cm, and a height 2.38 cm. (a) Calculate the area (trapezoid area = (upper base + lower base) $\times$ height $\div$ 2). (b) Approximate the area to the percentile. (c) First approximate each of the three lengths to tenths and then calculate the area. How does the result differ from (b)?		

V. Class afterhomework

Basic must-do questions (total 5 questions, must write determineprocess)

#	Question	Difficulty	Working Space
H1	Round 3.68 to the tenth place.		
H2	Approximate 9.053 to the percentile.		
H3	Approximate 8.172 to the tenth place.		
H4	A pen costs \$6.48 and a book costs \$3.75. How much is it worth in total? (Answers are approximated to tenths)		
H5	Estimate $6.3 \times 4.8 \approx \underline{\hspace{1cm}} \times \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$. Then calculate exactly $6.3 \times 4.8 = \underline{\hspace{1cm}}$. Are the two close?		

Advanced choose doquestion (total 3 questions,  choose do)

#	Question	Difficulty	Working Space
H6	Approximate 5.496 to tenths and hundredths and write the answers.		
H7	The rectangle is 3.87 m long and 2.45 m wide. Find the perimeter and area. Answers are approximated to the nearest tenth.		
H8	A number rounded to the percentile is 2.50. What is the range of values for this number? (i.e. What is the minimum possible? What is the maximum possible?)		

VI. The Lessoncorecommon errorssummary

✅ Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	Look at the wrong judgment position SEP : Take the tenth place but look at the tenth place yourself: Take the tenth place but look at the tenth	Take the nth position → Look at the n+1th position! For example, take the tenth place (1st

	place yourself	place) → look at the hundredth place (2nd place)
2	Braking early SEP : When taking tenths, stop only when you see the hundredths. Forgetting the thousandths may affect: When taking tenths, just stop when you see the hundredths. Forgetting the thousandths may affect	"Count one more digit before making a judgment" - always look at the next digit of the target position
3	The last zero is deleted SEP : $3.097 \approx 3.1$ (the last zero is missing): $3.097 \approx 3.1$ (missing the last zero)	$3.097 \approx 3.10$ (percentile), the last 0 must be retained to show accuracy
4	Confusing estimates with approximations	Estimate = Substitution calculation (before calculation); Approximation = Accurate result taken (after calculation)
5	Forget the chain carry when carrying 9 SEP : $3.297 \approx 3.30$ (percentile), not 3.29: $3.297 \approx 3.30$ (percentile), not 3.29	$9+1=10 \rightarrow$ Write 0 and advance 1 to the previous digit, check digit by digit.
6	Missing sentences/units in application questions	Must write "Answer: About _____ (unit)."
7	The results are different when approximating first and then calculating vs. calculating first and then approximating	Sub-tests usually require "accurate calculations first, and then approximate values". Pay attention to the question instructions.